

End to end Testing, Exploratory Testing

End to end testing

End-to-End (E2E) Testing validates the entire software workflow from the user's perspective, ensuring that the complete business logic functions correctly.

Core Process

- **User Simulation:** Testing the system as a real-world end user.
- **Scenario Identification:** Defining and executing diverse user journeys throughout the application.
- **E-commerce Example:** The full sequence from registration and product selection to payment and receipt of a confirmation SMS.

Key Challenges

- **Scenario Coverage:** Difficulty in identifying all possible user paths without extensive user feedback.
- **Environment Readiness:** Testing may be hindered if the environment lacks certain required functionalities.

Best Practices

- **Prototype-Based Cases:** Use user prototypes to develop realistic test scenarios.
- **Positive Focus:** Prioritize standard user interactions and successful flows over negative testing or system-breaking attempts.
- **Sequential Accuracy:** Execute test steps in the exact order a customer would follow.

Exploratory Testing

Exploratory Testing is an unstructured approach where testers investigate software without relying on prior documentation to discover defects that scripted testing might overlook.

Objectives

- **Discovery:** Identify new test cases and unconventional user paths.
- **Efficiency:** Provide rapid, early feedback when time for exhaustive testing is limited.
- **Documentation:** Record findings and observations during the exploration process.

Techniques

- **Time-boxing:** Restricting the exploration to a specific timeframe.
- **Targeted Goals:** Defining clear objectives for what to investigate.
- **Analysis:** Classifying bugs and reviewing results to refine future testing.